

EL YOUSFI HAJAR

EMBEDDED SYSTEMS & AI ENGINEERING STUDENT (M2)

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Availability : 6-month Final-Year Internship (PFE) starting February 2027 — Casablanca / Rabat

PROFESSIONAL SUMMARY

Master's student in ISERT engineering (Embedded Systems, Networks & Telecoms, FSTM), oriented towards embedded systems + AI: embedded vision (OpenCV, YOLOv8), ADAS, industrial AI and real-time IoT. Rigorous and passionate about innovation, seeking a 6-month Final-Year Internship (PFE) in embedded systems / artificial intelligence.

EDUCATION

FSTM – Master's in Embedded Systems, Networks and Telecommunications Engineering (ISERT) 2025 – 2027
• 2nd year (M2): embedded systems, networks & telecoms, image processing & vision, artificial intelligence.
FSTM – Bachelor's in Computer Science and Multimedia Networks (IRM) 2024 – 2025
FSTM – DEUST in Computer Science 2022 – 2024
Ain Harrouda High School – Baccalaureate in Physical Sciences (French Option) 2021 – 2022

INTERNSHIP

Proactive Load Balancing preserving QoE using Deep Learning (O-RAN 5G) | FSTM Jun – Aug 2026
• Predictive load-balancing xApp for the O-RAN Near-RT RIC: LSTM/GRU prediction (92% accuracy) + DRL agent (PPO), with handover decisions through the E2 interface (E2SM).
Design of a Smart Intrusion Detection Camera (Raspberry Pi, Embedded Deep Learning) | FSTM Jun – Aug 2026
• Real-time people detection on Raspberry Pi 4 (PFA): YOLOv8n, 3-level alerts, GPIO (LED/buzzer) and MQTT publishing.
MENPS – IT Directorate (Bachelor Internship) | Rabat Apr – Jun 2025
• Development of the Mostaqbali application (React.js, Spring Boot) for the Ministry of National Education: application management, tracking, validation, securing data flows and automating the administrative process.

PROJECTS

AI Safe Drive Guardian – Morocco-Localized Driving ADAS | Python, OpenCV 4, Dlib 19, YOLOv8n, Twilio 2026
• ADAS prototype (Driver Monitoring): drowsiness (EAR 91%), fatigue/yawning, phone use (93% at 37 FPS CPU) and distraction (head pose).
• Audio alerts in Darija, emergency SMS (Twilio) and real-time HUD; CLAHE preprocessing for high glare and night driving.
Smart Monitoring STEP OCP Khouribga – anomaly detection | Python, NumPy, Pandas, openpyxl 2026
• Isolation Forest + business thresholds on 171 daily measurements of an OCP water treatment station (volume, pH, turbidity): 60 anomalies flagged.
• Full pipeline: cleaning, normalization, scoring, enriched Excel export and interactive HTML dashboard.
WebSCADA Pro – Intelligent Industrial Monitoring | React.js, Node.js, Socket.io, Chart.js, InfluxDB, Python 2026
• Real-time web SCADA: dashboard, animated synoptic (SVG), 3-level alarms, 24h trends, OEE calculation and exportable reports.

TECHNICAL SKILLS

• Embedded systems : C, C++, GPIO, sensors, PROTEUS	• AI & data : Python, PyTorch, scikit-learn, Isolation Forest
• Embedded vision & deep learning : OpenCV, YOLOv8, Dlib, CNN, Raspberry Pi 4	• Development : React.js, Spring Boot, Java, SQL, Git, UML, Figma, C++ & embedded systems
• IoT & real-time : MQTT, WebSocket, FastAPI, SCADA, alarms	• Telecoms : MATLAB, CDMA chain, 2.4 GHz RF mixer

CERTIFICATIONS

• Claude Code in Action, Introduction to Model Context Protocol, Claude Code 101, Claude Platform 101, AI Fluency for Builders

LANGUAGES & ACTIVITIES

• **French** : Fluent, **English** : Fluent, **Arabic** : Native, **Italian** : Beginner, **Spanish** : Beginner, **Chinese** : Beginner
• **Activities** : Open Data Club (open data and analysis projects, FSTM 2026); Robotings Club (robotics and embedded electronics, workshops and tech competitions, FSTM 2023 – Present).